

PRERNA SRIVASTAVA

prerna7105@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [GeeksForGeeks](#) | Uttar Pradesh, India

EDUCATION

B.Tech — Computer Science | CGPA: 8.22 / 10

2023 – 2027

(Expected)

Deen Dayal Upadhyaya Gorakhpur University, Gorakhpur, Uttar Pradesh

Relevant Coursework: Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Database Systems, Object-Oriented Programming, Discrete Mathematics

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, SQL, HTML5, CSS3

Frameworks & Libraries: React.js, Node.js, Express.js, Tailwind CSS, scikit-learn, scikit-surprise, pandas, NumPy, Streamlit, Plotly

ML & Data Science: Supervised Learning, Collaborative Filtering, SVD Matrix Factorization, Ensemble Methods, Bayesian Regularization, Anomaly Detection, Explainable AI (XAI), Recommendation Systems

Mathematics: Probability & Statistics, Linear Algebra, Stochastic Gradient Descent, RMSE Optimization, Z-score, Descriptive Statistics

DSA: Arrays, Linked Lists, Trees, Graphs, Dynamic Programming, Greedy, Binary Search | 250+ problems on LeetCode & GFG

Databases & Tools: MongoDB, Firebase, SQL, Git, GitHub, REST APIs, Google Colab, Jupyter Notebook, Vercel

PROJECTS

Amazon Product Recommendation & Discovery Dashboard

github.com/PrernaSrivastava1/product-recommendation-system

- Built a multi-model recommendation system benchmarking User KNN, Item KNN, SVD Matrix Factorization, and Hybrid Ensemble on 7.8M Amazon reviews
- Engineered high-performance SVD model (RMSE: 0.883) via Stochastic Gradient Descent; resolved 99.27% matrix sparsity using latent factor spaces
- Implemented Bayesian Average regularization to solve cold-start problem; built Streamlit dashboard with XAI confidence breakdowns and NLP search
- Tech: Python, scikit-learn, scikit-surprise, Streamlit, Plotly, pandas, NumPy

AgriMitra — IoT Smart Agriculture System | agrimitra-sosk.onrender.com

- Applied threshold-based ML classification and rule-based decision logic on real-time IoT sensor streams (soil, temperature, humidity) for automated irrigation control
- Implemented Z-score anomaly detection for sensor noise filtering; built end-to-end time-series pipeline: ingestion → preprocessing → ML inference → dashboard

Campus Finance Hub — Trivia Pay | trivia-pay-new.vercel.app

- Applied anomaly detection and rule-based classification on transaction data to automate expense categorization; used descriptive statistics for real-time financial insights
- Tech: React.js, Node.js, JavaScript, Algorand Blockchain, Statistical Modeling

EXPERIENCE

AI Learning Intern | Samsung

Jul 2025 – Nov 2025

- Completed structured learning on Generative AI, LLM applications, model evaluation, and AI deployment workflows through Samsung's AI learning internship program
- Developed structured evaluation frameworks for AI outputs applying statistical thinking and iterative experimentation methodologies

Open Source Contributor | GirlScript Summer of Code (GSSoC)

Jul 2025 – Oct 2025

- Contributed features, resolved bugs, and improved documentation across multiple open-source repos using Git/GitHub in a nationally recognized program

RECOGNITION & AWARDS

CM of UP Award — Presented by Honourable Chief Minister Yogi Adityanath, Government of Uttar Pradesh

- Awarded for excellence in public communication and leadership at the state level

RIFT '26 Hackathon — Semi-Finalist & Top 3, Web3 + AI/ML Category

PhysicsWallah × Unstop | 8,000+ registrations | 700+ teams across 4 cities

- Led team of 3 — directed product architecture, ML + Web3 solution design, UI, and API planning within 24 hours